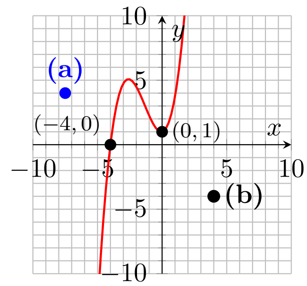
MTH 131(B) Dr. James Ashe
Exam 1 Friday 13$^{\text{th}}$ September, 2013 **Name:**

You must show your work to receive credit.

1. Use the graph to complete the following problems.



a. [3 pts] Graph and label the point $(-7.5, 4)$.

b. [3 pts] Determine the Cartesian coordinates of the point **(b)** on the graph.

$(4, -4)$

c. Use the graph to determine the $x-$int(s) if any of $f(x)$.

4, the point $(-4, 0)$.

d. Use the graph to determine the $y-$int(s) if any of $f(x)$.

1, the point $(0, 1)$.

Solve the linear equations.

2. $2x + 10 = 4$

$$\begin{aligned} 2x &= -6 \\ x &= -3 \end{aligned}$$

3. $3(x-1) + 13 = -(x+5)$

$$\begin{aligned} 3x - 3 + 13 &= -x - 5 \\ 4x &= -15 \\ x &= {}^{-15}\!/_4 \end{aligned}$$

4. $9x - (3 + 9x) = 9$

$$\begin{aligned} 9x - 3 - 9x &= 9 \\ -3 &\neq 9 \Rightarrow \text{No Solution} \end{aligned}$$

5. $\dfrac{3x}{2} - x = \dfrac{x}{10} - \dfrac{8}{5}$

$$\begin{aligned} 10\left(\frac{3x}{2} - x\right) &= \left(\frac{x}{10} - \frac{8}{5}\right)10 \\ 15x - 10x &= x - 16 \\ 4x &= -16 \\ x &= -4 \end{aligned}$$

6. $\dfrac{x+5}{3} = \dfrac{5}{3} + \dfrac{x-8}{2}$

$$\begin{aligned} 6\left(\frac{x+5}{3}\right) &= 6\left(\frac{5}{3} + \frac{x-8}{2}\right) \\ 2(x+5) &= 10 + 3(x-8) \\ 2x + 10 &= 10 + 3x - 24 \\ -x &= -24 \Rightarrow x = 24 \end{aligned}$$

Identify any restrictions on the variable, and solve the rational equation.

7. $\frac{2}{3} = \frac{4}{x} + 1$

$$x \neq 0$$

$$\begin{aligned} 3x\left(\frac{2}{3}\right) &= 3x\left(\frac{4}{x}+1\right) \\ 2x &= 12+3x \\ -12 &= x \Rightarrow x = -12 \end{aligned}$$

Solve the quadratic equations by factoring.

8. $x^2 - 2x = 8.$

$$\begin{aligned} x^2 - 2x - 8 &= 0 \\ (x-4)(x+2) &= 0 \\ &\Rightarrow x = 4 \text{ and } x = -2 \end{aligned}$$

9. $4x^2 - 8x - 32 = 0.$

$$\begin{aligned} 4(x^2 - 2 - 8) &= 0 \\ 4(x-4)(x+2) &= 0 \\ &\Rightarrow x = 4 \text{ and } x = -2 \end{aligned}$$

Solve the quadratic equation using the square root property.

10. $(x-3)^2 - 5 = 7$

$$\begin{aligned} (x-3)^2 &= 12 \\ \sqrt{(x-3)^2} &= \pm\sqrt{12} \\ x - 3 &= \pm\sqrt{4}\sqrt{3} \\ x &= 3 \pm 2\sqrt{3} \end{aligned}$$

Solve the quadratic.

11. $x^2 + 2x = 13.$

By completing the square. $b = 2 \Rightarrow \left(\frac{2}{2}\right)^2 = 1$

$$\begin{aligned} x^2 + 2x + 1 &= 13 + 1 \\ (x+1)^2 &= 14 \\ x + 1 &= \pm\sqrt{14} \\ x &= -1 \pm \sqrt{14} \end{aligned}$$

Using the quadratic formula; $x^2 + 2x - 13 = 0$; $a = 1$, $b = 2$, and $c = -13$.

$$\begin{aligned} x &= \frac{-(2) \pm \sqrt{(2)^2 - 4(1)(-13)}}{2(1)} \\ &= \frac{-2 \pm \sqrt{56}}{2} = \frac{-2 \pm 2\sqrt{14}}{2} \\ &= -1 \pm \sqrt{14} \end{aligned}$$

Solve the problems

12. [7 pts] When three times a number is subtracted from eight times the same number the result is 20. What is the number?

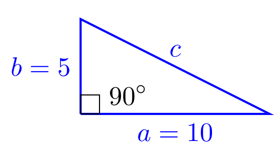
$$\begin{aligned} 8x - 3x &= 20 \\ 5x &= 20 \\ x &= 4 \end{aligned}$$

13. [7 pts] John buys house for $100,000. If the house increases in value by $3500 per year. How long until it doubles in value (i.e., how long until it is worth $200,000)?

$$\begin{aligned} 100,000 + 3500x &= 200,000 \\ 3500x &= 100,000 \\ x &\approx 28.57 \text{ years} \end{aligned}$$

14. [Bonus] Jane is playing playing receiver on a flag football team. She runs 10 yards straight down-field, makes a 90°turn to her right, and runs another 5 yards straight towards the sideline. How far is she from her original position.

The distance is the hypotenuse of a right triangle.



Use Pythagoras' theorem to solve for c,

$$\begin{aligned} c^2 &= 10^2 + 5^2 \\ c &= \sqrt{125} \text{ yards} \end{aligned}$$